LATEX WRITESENSE AI

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Main Project

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Outline

Introduction

- Academic and technical writing requires accuracy, clarity, and coherence.
- Overleaf is widely used for LaTeX-based research papers and theses.
- However, Overleaf provides limited support for writing-quality improvement.
- Most AI writing tools operate outside the LaTeX environment, disrupting workflow and risking syntax errors.
- This project introduces a selection-based AI writing assistant integrated directly into Overleaf.
- The system provides grammar correction, clarity analysis, summarization, multilingual translation, and research paper identification.

Introduction

Objectives:

The primary objective of this project is to integrate an AI-assisted writing support system directly into the Overleaf environment to enhance academic writing efficiency.The system preserves the LaTeX syntax integrity while ensuring full user control over all suggested modifications.

Literature Review

J. Hou, A. L. Huikai, N. Chen, Y. Gong, and B. He, "PaperDebugger: A Plugin-Based Multi-Agent System for In- Editor Academic Writing, Review, and Editing,"

- Integrated AI assistance directly into Overleaf
- Supported selection-based in-editor editing
- Provided LLM-driven writing and review support
- Eliminated external copy–paste workflows
- **Observation:** Shows the benefit of in-editor AI writing tools, but uses complex multi-agent architectures, motivating simpler and more controlled designs. [1]

Literature Review

S. Totawad, R. Patil, A. Thorat, D. Sasane, Y. D. Sinkar and S. Gaikwad, "Research Papers Recommendation System Prediction Using Deep Learning and LLMs,"

- Proposed an AI-based research paper recommendation system
- Used deep learning and LLM embeddings for semantic similarity
- Recommended papers based on content, citations, and user preference
- Improved research discoverability in large academic repositories
- **Observation:** Demonstrates the effectiveness of LLM-based paper recommendation, but operates as a standalone system, not integrated into the LaTeX writing workflow.[2]

Literature Review

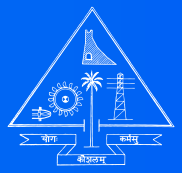
Jovic, M., Papakonstantinidis, S. Kirkpatrick, R. "From redink to algorithms: investigating the use of large language models in academic writing feedback"

- Compared human feedback vs. LLM-generated feedback.
- Evaluated grammar, structure, tone,clarity in student writing.
- Showed that human feedback excels in contextual understanding, deeper reasoning, and personalized guidance
- Demonstrated that LLMs provide consistent and efficient surface-level feedback
- Reported limitations in contextual understanding and personalized guidance
- **Observation:** Highlights the need for controlled, transparent AI feedback

Literature Review Summary

Paper	Focus Area	Key Contributions and Observations
Hou et al. (2023) [1]	In-editor AI writing support	Integrated AI assistance directly into Overleaf with selection-based editing. Provided LLM-driven writing and review support while eliminating external copy–paste workflows.
Totawad et al. (2024) [2]	Research paper recommendation	Proposed an AI-based paper recommendation system using deep learning and LLM embeddings for semantic similarity. Recommended papers based on content, citations, and user preferences.
Jovic et al. (2023) [**?**]	LLM-based academic feedback	Compared human feedback with LLM-generated feedback across grammar, structure, tone, and clarity. Found that humans excel in contextual understanding and personalized guidance, while LLMs provide consistent and efficient surface-level feedback.

Table 1: Summary of Related Work



Existing Systems

- Overleaf mainly focuses on LaTeX document formatting and collaboration.
- No built-in support for grammar correction or clarity analysis.
- No readability evaluation or academic writing assistance.
- Dependence on external writing tools requiring manual copy–paste of LaTeX content.
- Workflow disruption when switching between the editor and external tools.
- Risk of LaTeX syntax breakage (commands, equations, references) during external processing.
- Lack of integrated features such as summarization, translation, and research paper identification.

Proposed System

- Provides AI-assisted writing support directly within the Overleaf LaTeX editor.
- Eliminates manual copy–paste by integrating writing assistance into the editor.
- Performs grammar correction using a rule-based approach and AI-based.
- Analyzes writing clarity using readability metrics and AI evaluation.
- Generates summaries of selected text to support quick content understanding.
- Enables multilingual translation of academic text.
- Recommends relevant research papers using keyword-based scholarly search.
- Preserves LaTeX syntax and applies changes only with user approval.

Requirements Specification

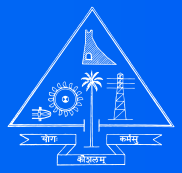
Hardware Requirements

- **Memory:** Minimum 8 GB RAM
- **Processor:** Intel Core i5 / AMD Ryzen 5 or higher
- **Storage:** Minimum 20 GB free HDD/SSD space
- **Internet Connectivity:** Required for accessing the Overleaf editor and retrieving research papers from scholarly databases

Requirements Specification

Software Development Requirements

- **Programming Language:** Python 3.10
- **Framework:** FastAPI
- **Backend Server:** Uvicorn
- **Grammar Checking Tool:** LanguageTool
- **AI Model:** LLaMA 3
- **Translation Model:** NLLB (facebook/nllb-200-distilled-600M)
- **Research Paper Retrieval:** CrossRef API



Requirements Specification

Functional Requirements

- Processes only user-selected sentences or paragraphs.
- Performs grammar correction while preserving the original meaning.
- Analyzes writing clarity using readability metrics and AI evaluation.
- Provides clarity feedback and improvement suggestions.
- Summarizes selected text without introducing new information.
- Supports multilingual translation while preserving LaTeX syntax.
- Identifies relevant research papers based on selected text.

Non-Functional Requirements

- **Performance:** Provides acceptable response time for grammar correction, clarity analysis, and other text processing tasks.
- **Scalability:** Supports future extension with additional writing features and language models.
- **Security & Privacy:** Ensures that the user's academic content is handled safely through controlled backend communication."

Input and Output

Input

- User-selected text from a LaTeX document in the Overleaf editor

Output

- Corrected text with grammatical errors fixed
- Clarity analysis with readability metrics and AI feedback
- Concise summary of the selected text
- Translated text in the user-specified target language
- List of relevant research papers related to the selected text

System Architecture

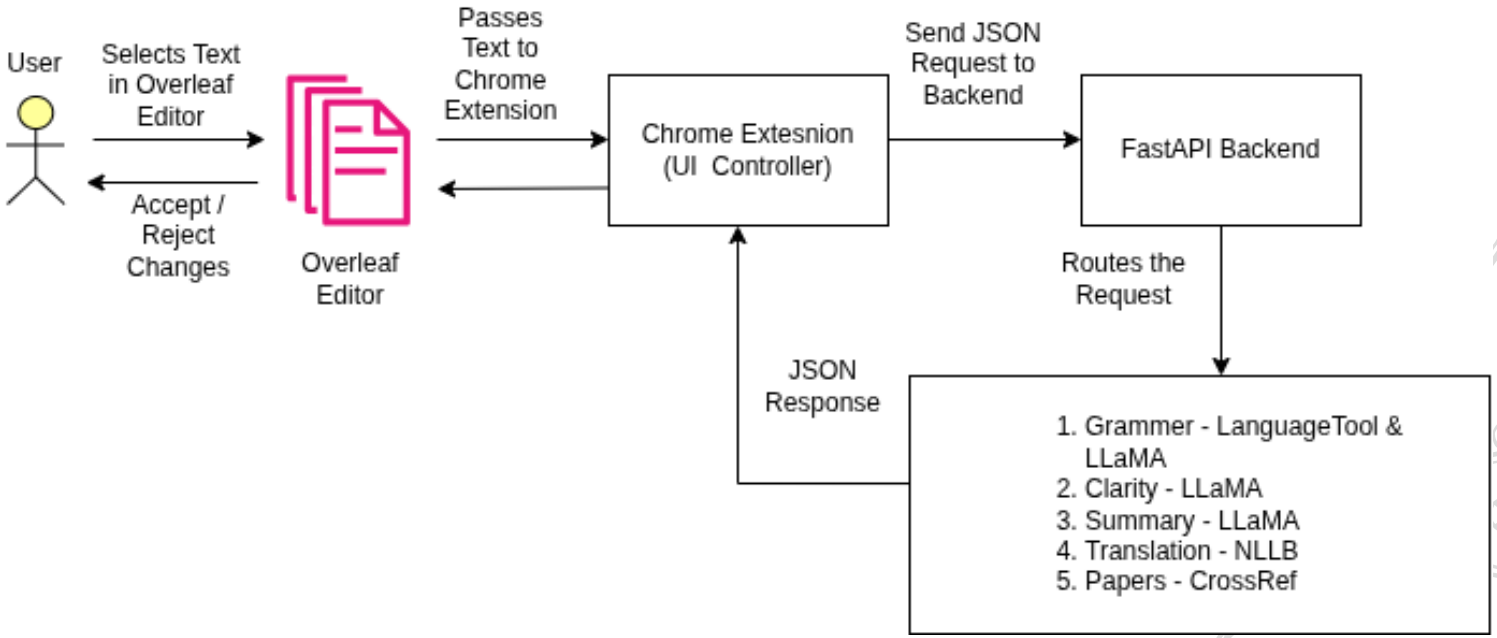


Figure 1: Architecture Diagram

Data Flow Diagrams

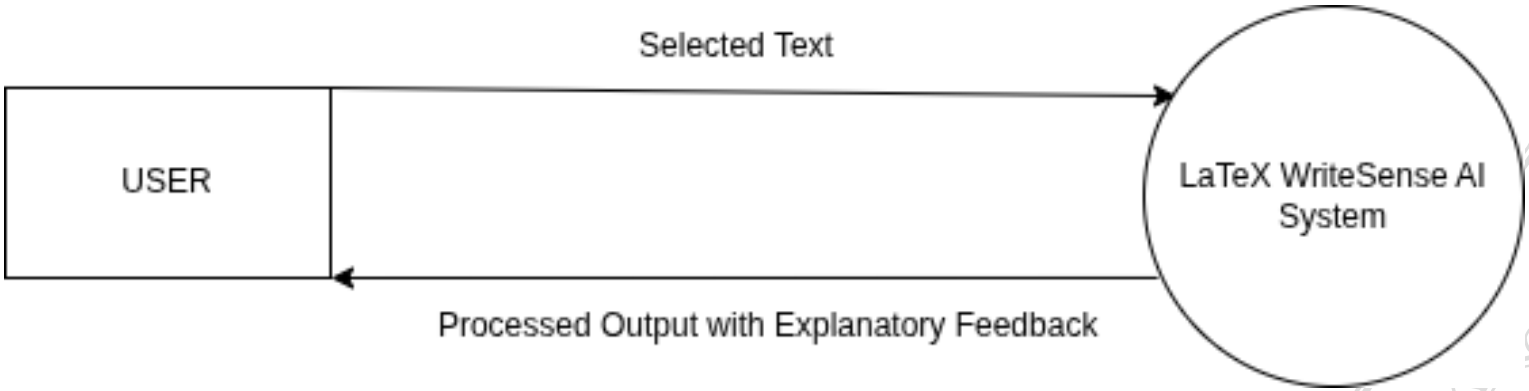


Figure 2: Level 0 DFD

Data Flow Diagrams

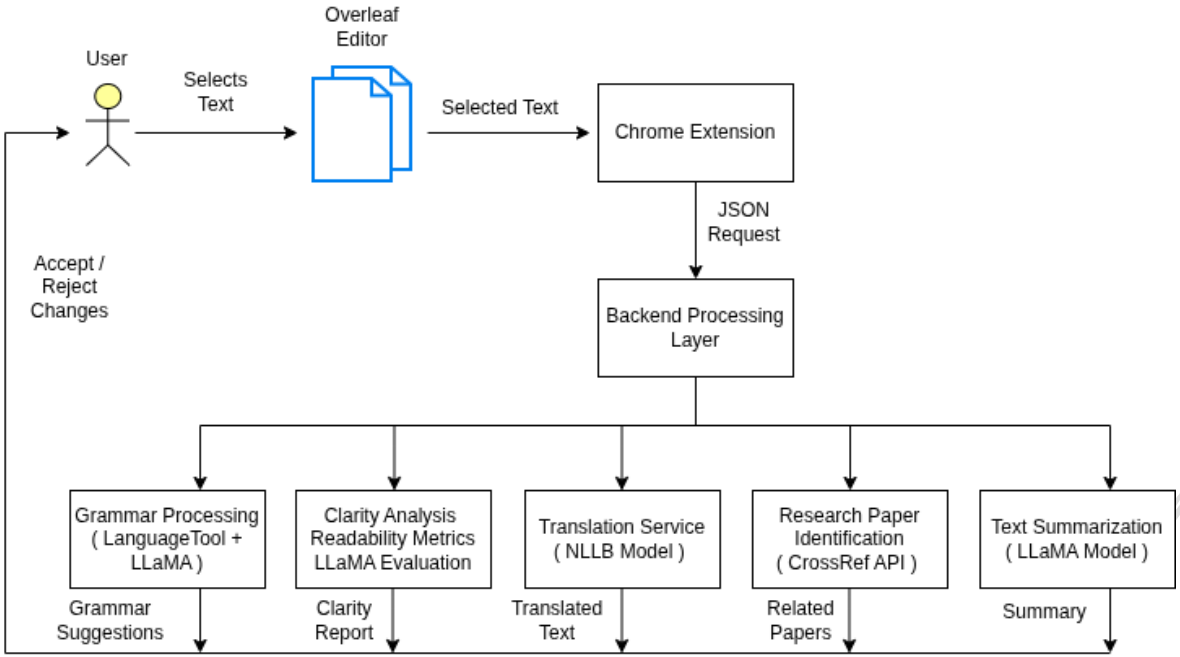
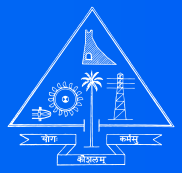


Figure 3: Level 1 DFD

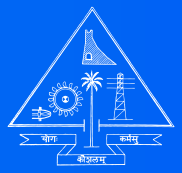
System Modules

- **User Interface Module:** Provides user interaction through a Chrome extension integrated with the Overleaf LaTeX editor.
- **Text Selection & Input Handling Module:** Captures user-selected sentences or paragraphs from the LaTeX editor and prepares the text for processing.
- **Backend Request Management Module:** Receives requests from the extension, validates inputs, and routes them to appropriate backend services using FastAPI.
- **Grammar Correction Module:** Performs AI-Based,Rule-based grammatical analysis using LLaMA,LanguageTool and generates safe corrections without affecting LaTeX syntax.



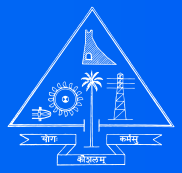
System Modules

- **Clarity Analysis Module:** Evaluates academic writing quality using readability metrics and AI-based structural evaluation.
- **Summarization Module:** Generates concise summaries of selected academic text while preserving the original meaning.
- **Multilingual Translation Module:** Translates selected text into the target language while preserving LaTeX commands and structure.
- **Research Paper Identification Module:** Extracts research-related keywords and retrieves relevant academic papers from scholarly databases.
- **Result Presentation & User Control Module:** Displays results and suggestions to the user and allows changes to be applied only after user approval.



Implementation

- Successfully implemented the core processing pipeline of the **Write-Sense AI academic writing assistant** integrated with the Overleaf LaTeX editor.
- **Text Selection and Input Capture:** The Chrome extension captures user-selected text from the Overleaf editor and prepares it for processing.
- **Backend Request Routing:** Selected text is sent to FastAPI backend services where requests are validated and routed to appropriate processing modules.
- **Grammar Processing:** AI-Based,Rule-based grammar checking is performed using LLaMA and LanguageTool to detect errors and generate safe corrections without affecting LaTeX syntax.



Implementation (contd.)

- **Clarity Analysis:** Academic readability and structural quality are evaluated using metrics such as Flesch Reading Ease, lexical density, clause density, and passive voice ratio.
- **Summarization and Translation:** Selected text can be summarized or translated into multiple languages using transformer-based models while preserving LaTeX structure.
- **Research Paper Identification:** Relevant academic papers are retrieved from cross-ref database based on extracted keywords and research domain analysis.
- **Result Presentation:** All generated outputs are displayed in a dock-style interface within the Chrome extension, allowing the user to review and optionally apply the suggested changes.

Final Results

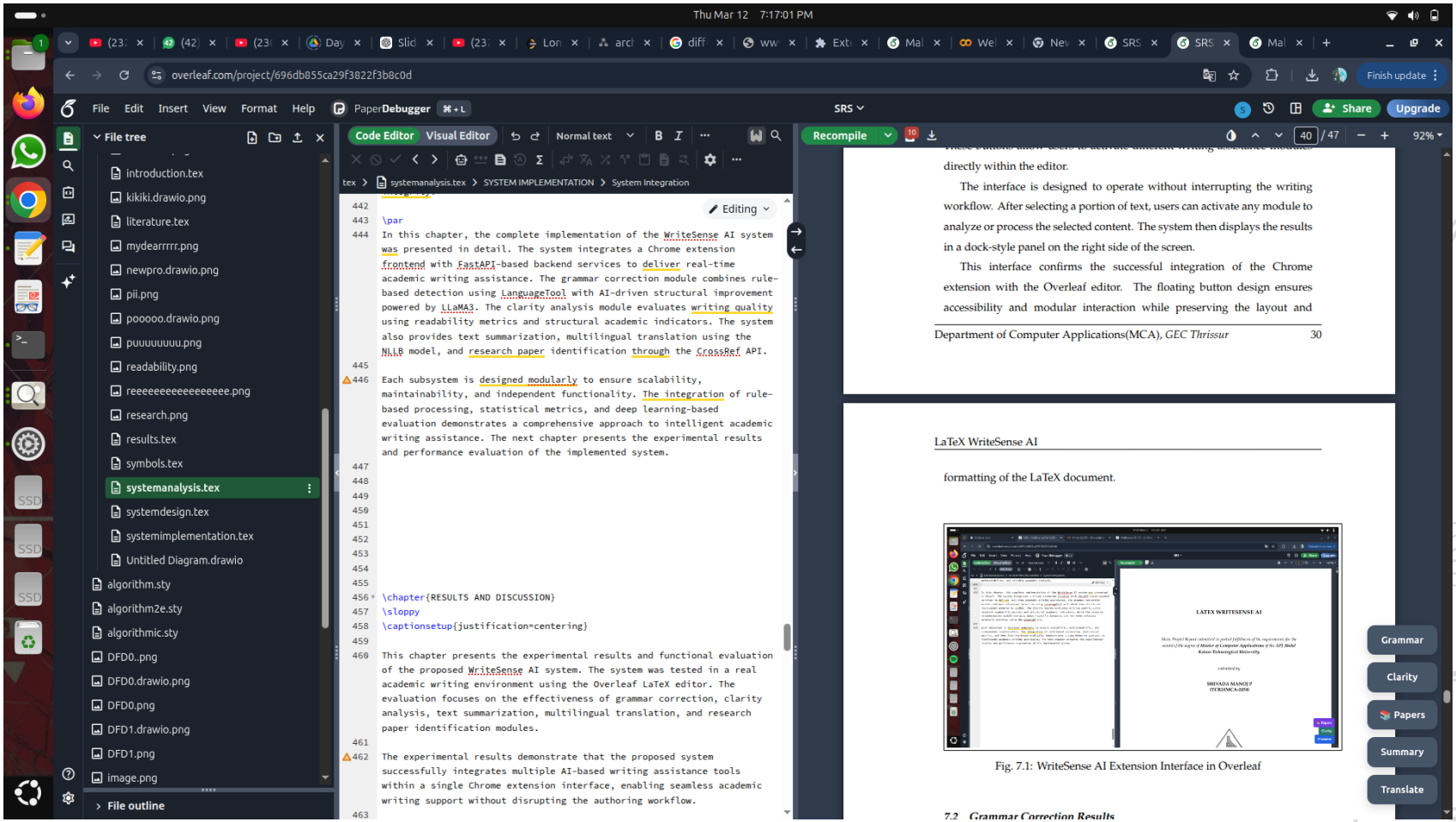


Figure 4: Interface

Final Results

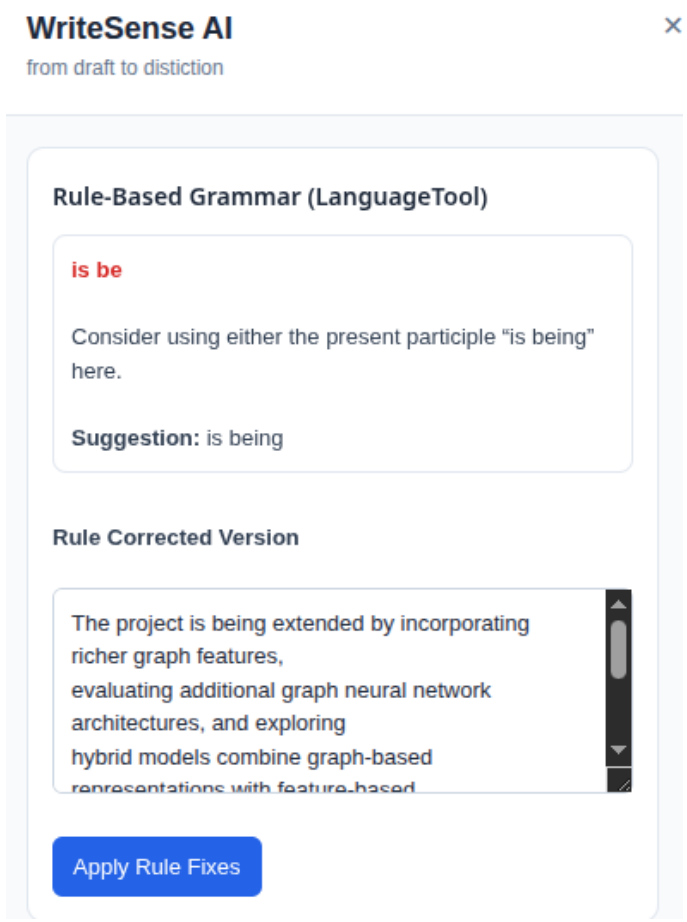


Figure 5: Grammar Correction

Final Results

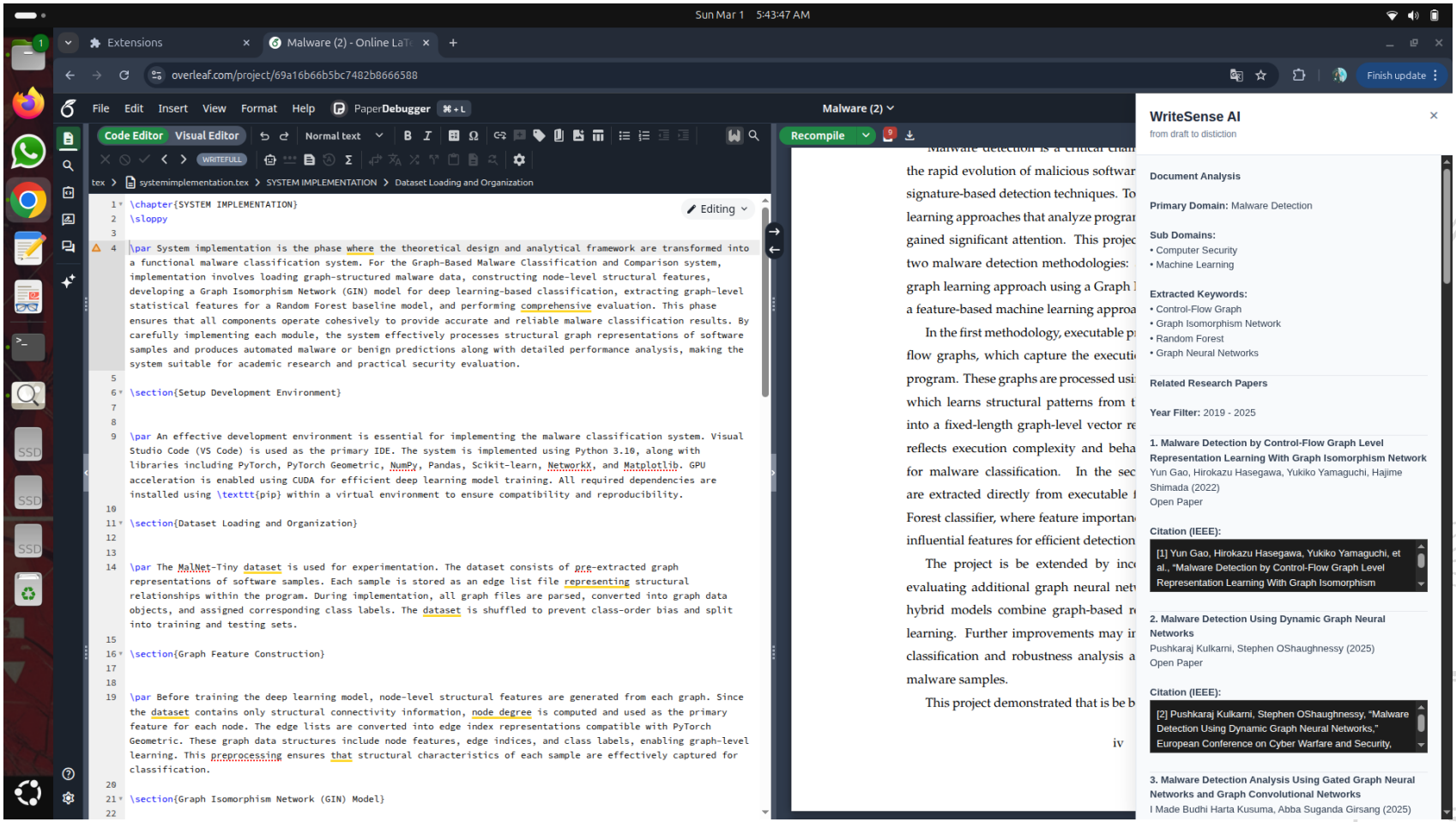


Figure 6: Research Paper Recommendation

Final Results

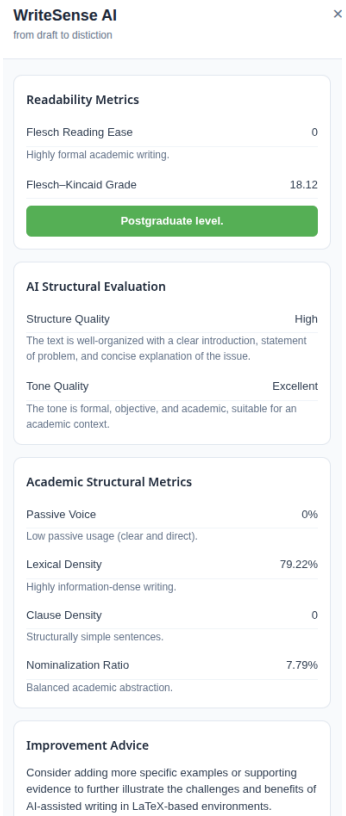


Figure 7: Clarity Analysis

Final Results

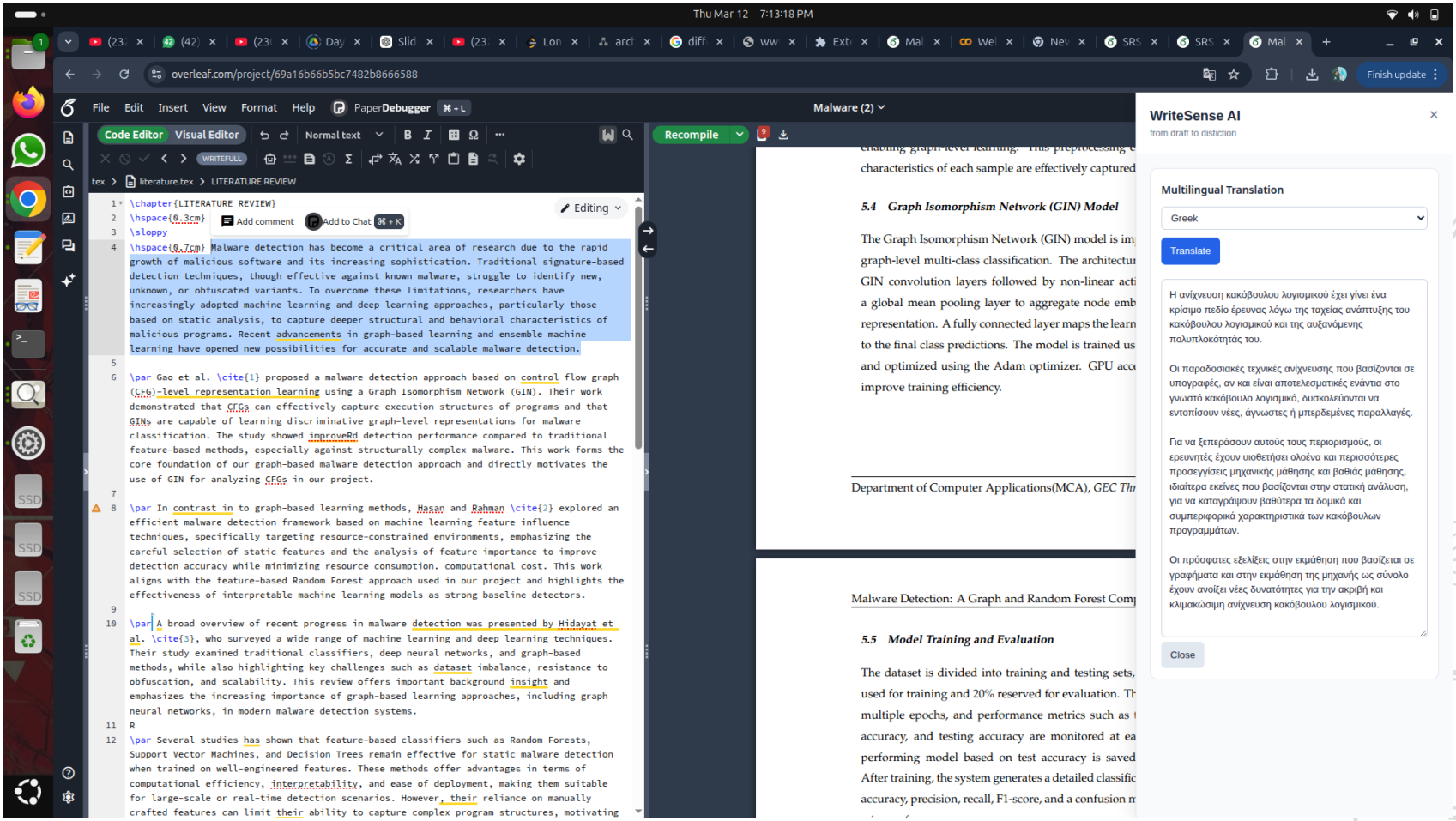


Figure 8: multilingual Translation

Final Results

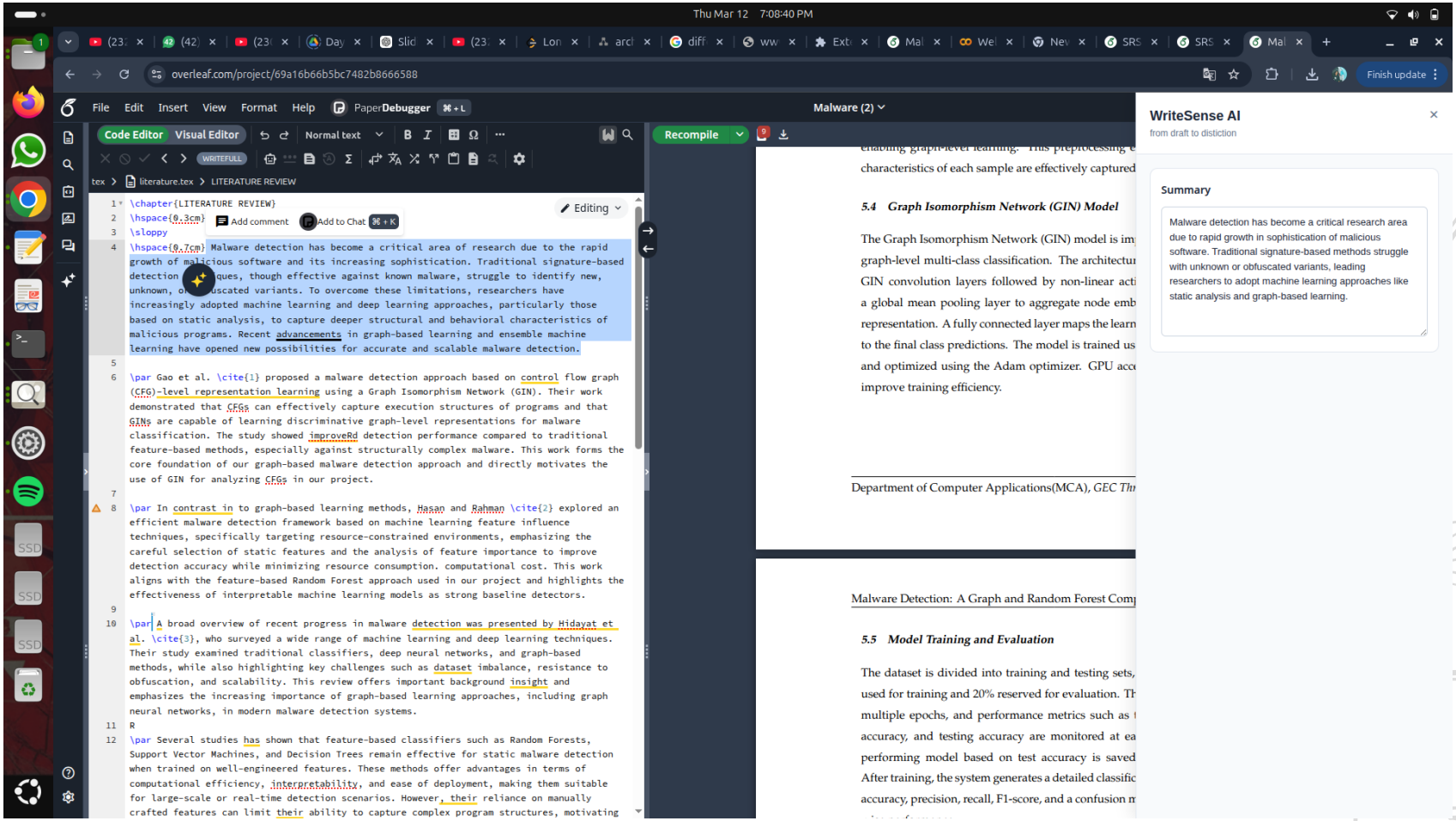
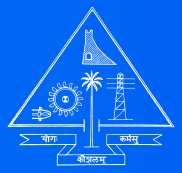


Figure 9: Summarization

Future Enhancements

- Use advanced language models to improve grammar correction and make writing clearer.
- Provide citation suggestions to help users find relevant references while writing.
- Improve system performance to make responses faster and more efficient.
- Enable paraphrasing for better academic writing
- Recommend LaTeX commands to help users format documents correctly
- Expand platform support beyond Overleaf



Conclusion

- Successfully developed **WriteSense AI**, an AI-assisted academic writing support system integrated with the Overleaf LaTeX editor through a Chrome extension.
- Implemented a modular architecture combining a Chrome extension frontend with **FastAPI-based backend services** for text processing.
- Integrated rule-based grammar correction using **LanguageTool** and AI-based using LLaMA.
- Provided additional writing assistance features including **summarization, multilingual translation, and related research paper identification**.
- Designed the system to process only **user-selected text**, ensuring transparency, LaTeX syntax preservation, and full user control over document modifications.

References

[1] J. Hou, A. L. Huikai, N. Chen, Y. Gong, and B. He, "PaperDebugger: A Plugin-Based Multi-Agent System for In-Editor Academic Writing, Review, and Editing," arXiv preprint arXiv:2512.02589, 2025.

[2] S. Totawad, R. Patil, A. Thorat, D. Sasane, Y. D. Sinkar and S. Gaikwad, "Research Papers Recommendation System Prediction Using Deep Learning and LLMs," 2025 International Conference on Computing Technologies (ICOCT), Bengaluru, India, 2025

[3] Jovic, M., Papakonstantinidis, S. & Kirkpatrick, R. "From red ink to algorithms: investigating the use of large language models in academic writing feedback" Lang Test Asia 15, 59 (2025). https://doi.org/10.1186/s40468-025-00389-2.

Thank You